Class - 11th
Chapter- Set Theory
Hand Written Notes

Chapter- Sets

Class- 11th

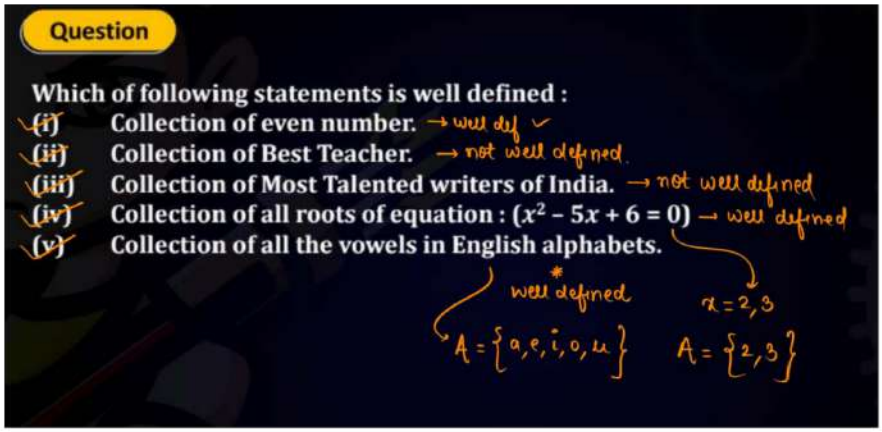
Definition of set

A set is well defined collection of distinct object. Sets are usually denoted by capital letters A, B, C, D, X, Y.

Note!- The element of the set are denoted by small letter. If x is an element of a set A, we write $x \in A$ (read as x belongs to A')

If x is not an element of A, then we write $x \notin A$ (read as x does not belongs to A').

Ex → If $A = \{1, 3, 5, 6\}$, then $3 \in A$, $6 \notin A$



Note!- Cardinal number of a Set
or
Order of a Set
or
Number of element in a set
$n(A)$ = Number of element in set A

$A = \{1, 2, 5\}$
Cardinal no = 3
Order of set A = 3
$n(A) = 3$

Representation of Sets:-

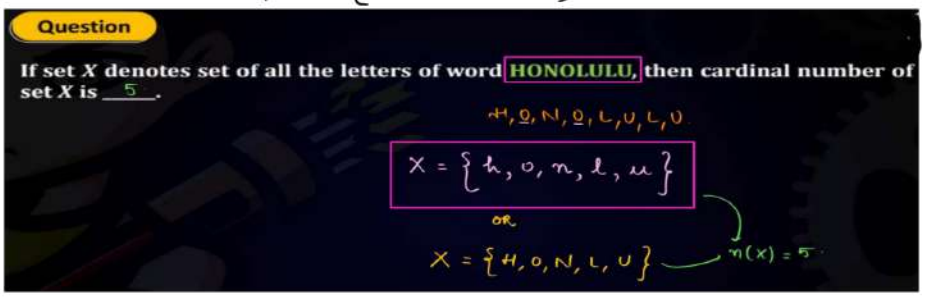
2. Roaster form or Tabular form :-

Structure : $A = \{ , , , , \dots \}$ => $A = \{ _ , _ , _ \}$ (element)

Note!-
i) Repetition of element (number) is not allowed.
ii) Order of element is not important.

Ex → Form a set of all the "vowels"

$X = \{a, e, i, o, u\}$ => $n(X) = 5$



Important Symbols :-

	Symbol	meaning
①	: or |	Such that
2)	$\in$	belongs to
3)	$\notin$	does n't belongs to

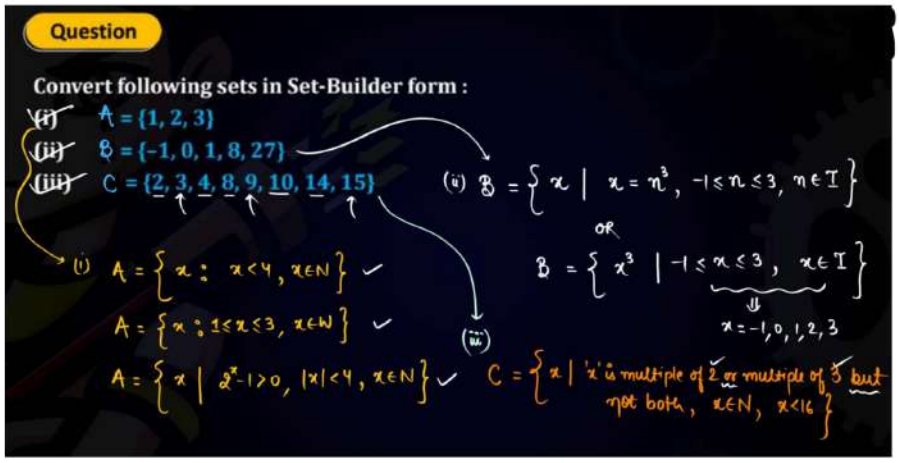
2) Set-Builder form :-

"Element are not directly visible.

$$A = \{x \mid \text{properties satisfied by } x\}$$

Ex → $A = \{x \mid x \in N,\ x < 3\} = \{1, 2\}$

$X = \{\alpha \mid \alpha \text{ is a prime no}, \alpha^2 - 11\alpha + 28 = 0\} = \{7\}$



Types of Sets :-

1) Empty Set / Null set / Void set :-

Denoted by → $\phi = \{\}$

Note : $\{0\} \neq \phi$

Note : $n(\phi) = 0$

2) Finite Set :-

Jis mein element countable ho

Ex : $X = \{x \mid x^2 - 5x + 6 = 0\} = \{2, 3\}$ → finite set

3) Inifinte Set :-

Set having infinite elements.

Ex → N, W, Q, R

Ex → $X = \{x \mid 2 \leq x < 4, x \in R\}$

4) Singleton Set :-

Set having 1 element only.

if $n(A) = 1$ ⇒ 'A' is singleton set.

5) Equivalent Set :-

Two set having equal number of element

If $n(A) = n(B)$ $\Rightarrow$ A & B are equivalent.

Ex → $A = \{1, 2\}$

$B = \{a, b\}$

6) Equal Set :-

Two set in which number of element are equal & element are also same.

If $A = B$ $\Rightarrow$ Then A & B have same & equal elements.

Ex → $A = \{1, 2, 3\}$

$B = \{x \mid x^x - 1 > 0,\ |x| < 4,\ x \in N\}$

Subsets :-

Let A and B be two set if every element of A is an element of set B, then A is called a subset of B.

Note :- If set A is subset of set B then B is called Super Set of set A.

Ex → $A = \{1, 2, 3, 4\}$

$B = \{1, 2, 3, 4, 6, 7, 8\}$

→ A is subset of B → $A \subset B$

↳ B is superset of A.

Proper Subset :- If set A is subset of B and A & B are not equal then A is called proper subset of set B

> no of proper set = no of subset − 1

Important Points :-

→ Empty set is a subset of every set

→ Every Set is a subset of itself $A \subset A$

→ If A is a subset of B and B is a subset of A then $\Leftrightarrow$ $A = B$

> # Number of subset of any set $A = 2^{n(A)}$
>
> # Number of proper subsets $= 2^{n(A)} - 1$

Question

Consider set $A = \{1, 2, 3, 4, 5\}$, then find

(i) Number of subsets of set A $= 2^5 = 32$.

(ii) Number of proper subsets of set A $= 2^5 - 1 = 31$.

(iii) Number of non-empty subsets of set A

jo khali na ho → $\phi = \{\}$

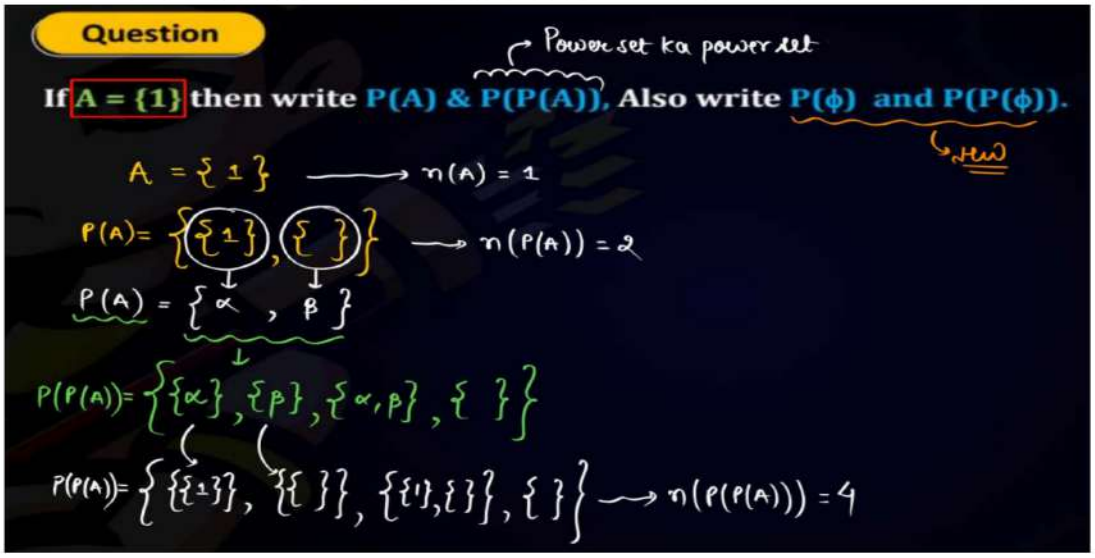
$2^5 - (1) = 31$.

↳ ϕ ko hatana hai.

Power Set :-

The collection of all subset of a set A is called the power set of A. It denoted by P(A).

$n(P(A)) =$ number of element in power of set of A = No of subset of set A $= 2^{n(A)}$



Universal Set :-

A set consisting of all element which occur in the discussion in called a universal and is denoted by U.

Venn diagram :-

Geometrical representation of sets.

U = Universal ⇒

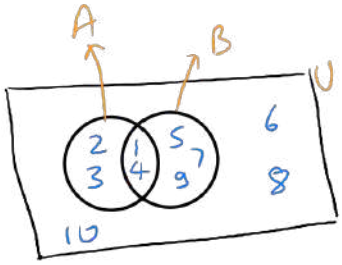


any set

Ex → A = {1, 2, 3, 4}

B = {1, 4, 5, 7, 9}

U = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}



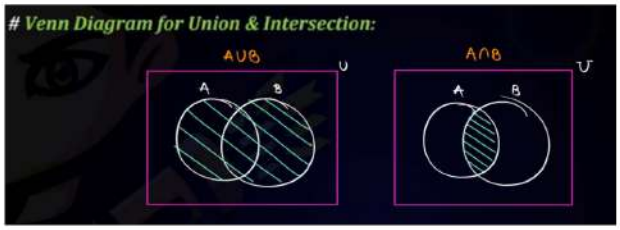
Operation of two Sets :-

Union :-

A ∪ B = (combined element of set A & set B)

Intersection :-

A ∩ B = (common element of set A & set B)



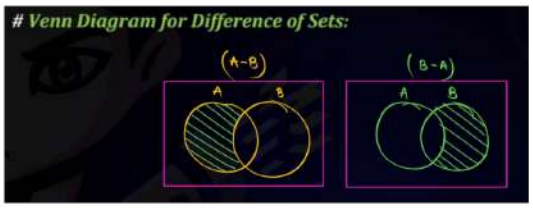
Difference of Sets :-

$A - B = A - (A \cap B)$

"Set A mein se A & B ke common element hta do

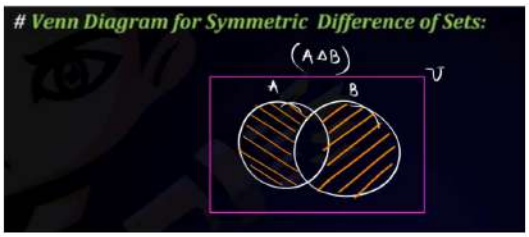
$B - A = B - (A \cap B)$

Set B meinse A & B ke common element hta do



Symmetric Difference :-

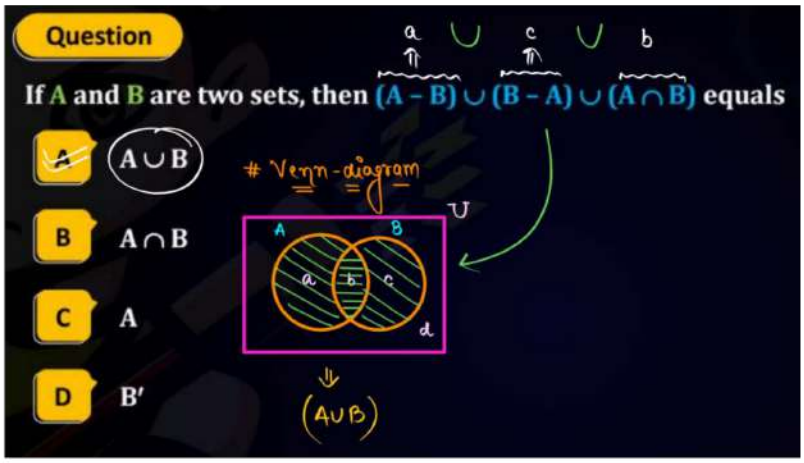
Symbol :- $\triangle$ & $\oplus$ $\quad A \triangle B = A \oplus B = (A-B) \cup (B-A)$



Complement :-

denoted by : $\overline{A} / A' / A^c \equiv$ Complement of set 'A' = not A

$$\boxed{\overline{A} = U - A}$$



De-Morgan's Law :-

1)	$(\overline{A \cup B}) = \overline{A} \cap \overline{B}$
2)	$(\overline{A \cap B}) = \overline{A} \cup \overline{B}$

Law of Algebra :-

Idempotent Laws :-

a) $A \cup A = A$ $\quad$ b) $A \cap A = A$

Identity Laws :-

a) $A \cup \phi = A$ $\quad$ b) $A \cap U = A$ $\quad$ c) $A \cap \phi = \phi$

Commutative Laws :-

a) $A \cup B = B \cup A$ $\quad$ b) $A \cap B = B \cap A$

c) $A \triangle B = B \triangle A$

Associative Law :- If A, B and C are any three sets, then

a) $(A \cup B) \cup C = A \cup (B \cup C)$ b) $(A \cap B) \cap C = A \cap (B \cap C)$

Distributive law :-

a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

b) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Important formula's

for two sets

1) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

2) If A and B are disjoint set then $n(A \cup B) = n(A) + n(B)$

3) $n(A - B) = n(A) - n(A \cap B)$

4) $n(A \triangle B) = n(A \cup B) - n(A \cap B)$

for three sets

v) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

